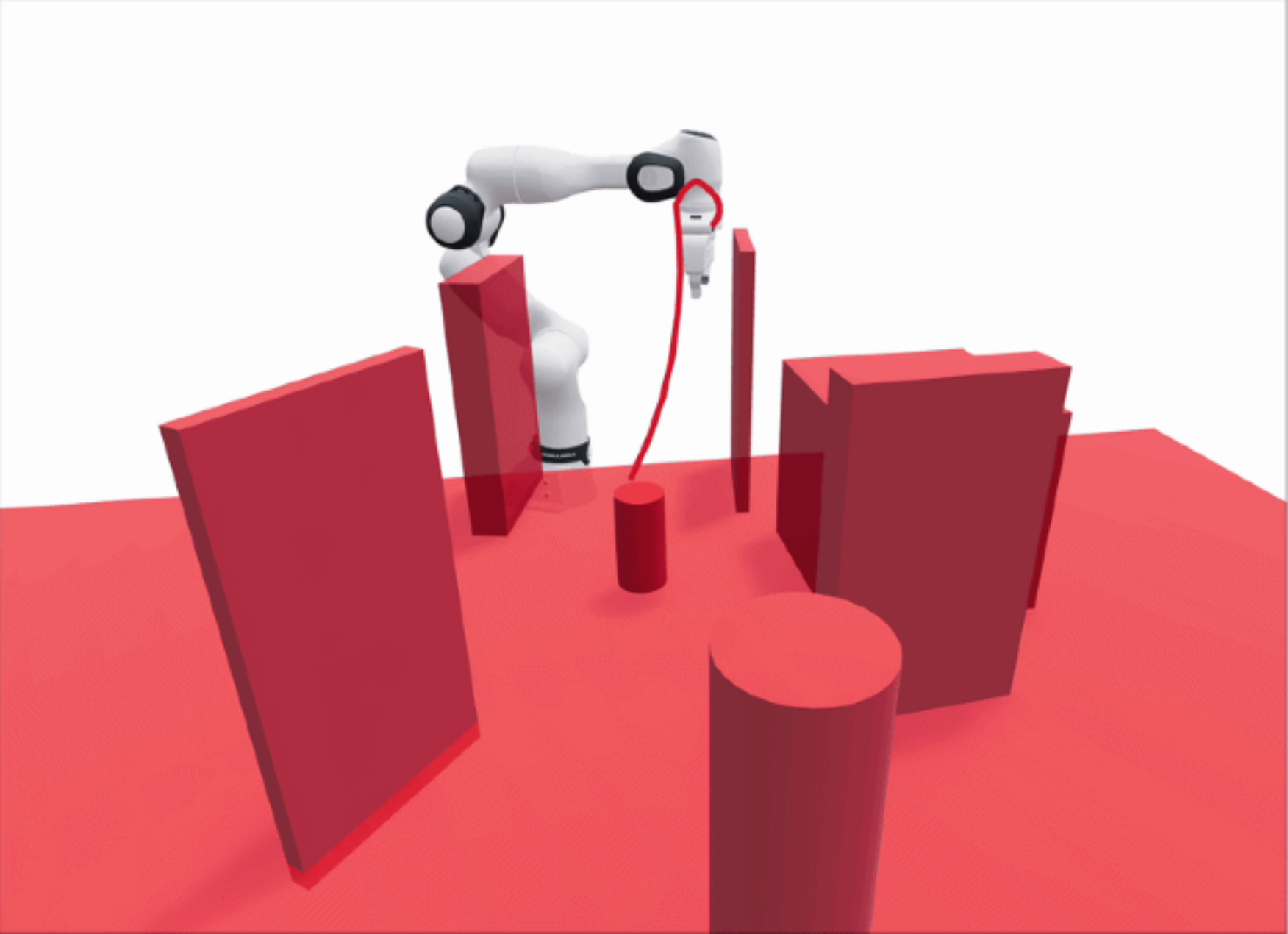
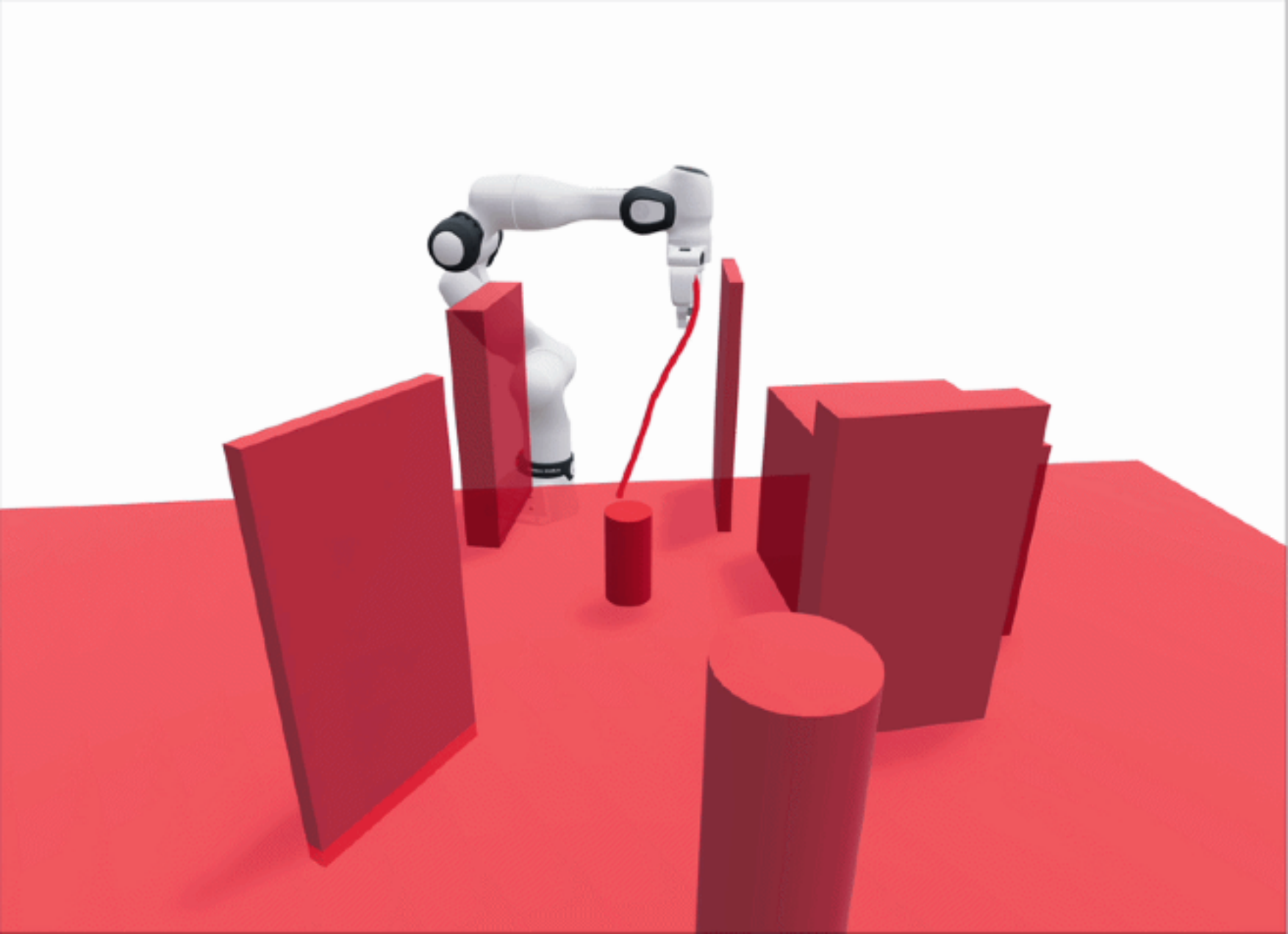






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joint-space motion that respect inertia

Kinetic-energy metric

Evidence and metrics

joint-space straight-line motions

Optimization baselines often need hand-tuned barriers to avoid obstacles

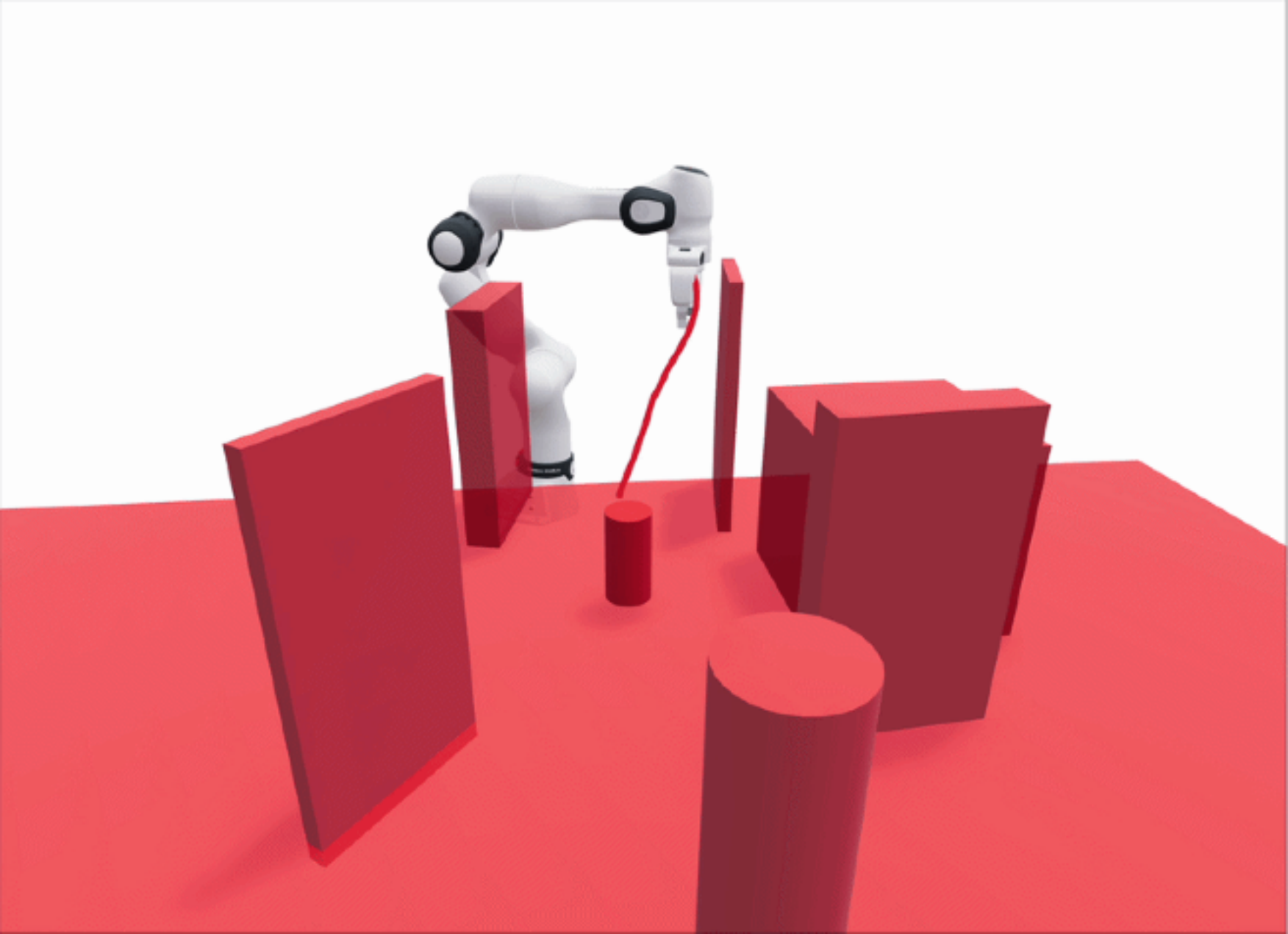
— and can still get stuck in local minima.

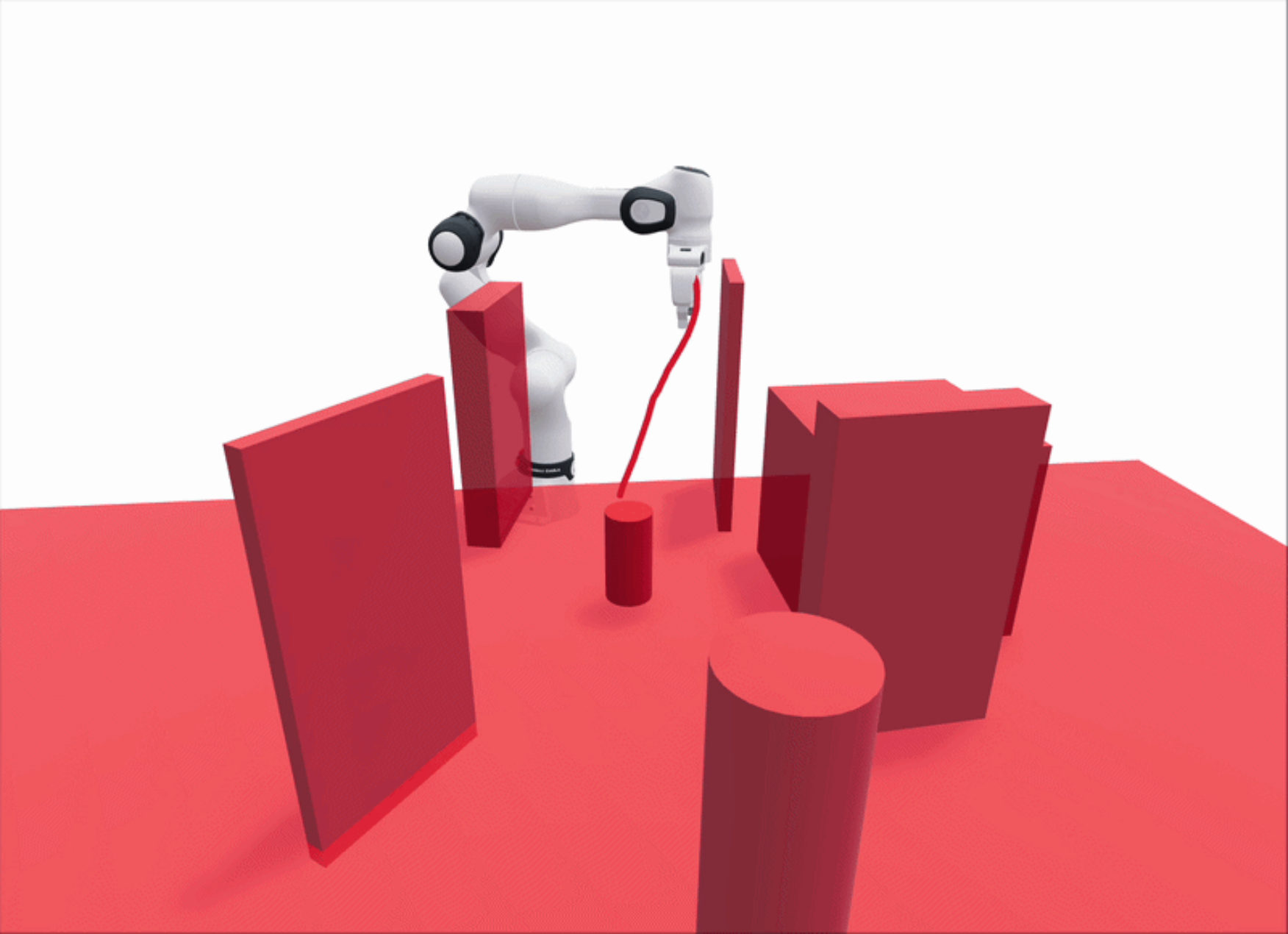


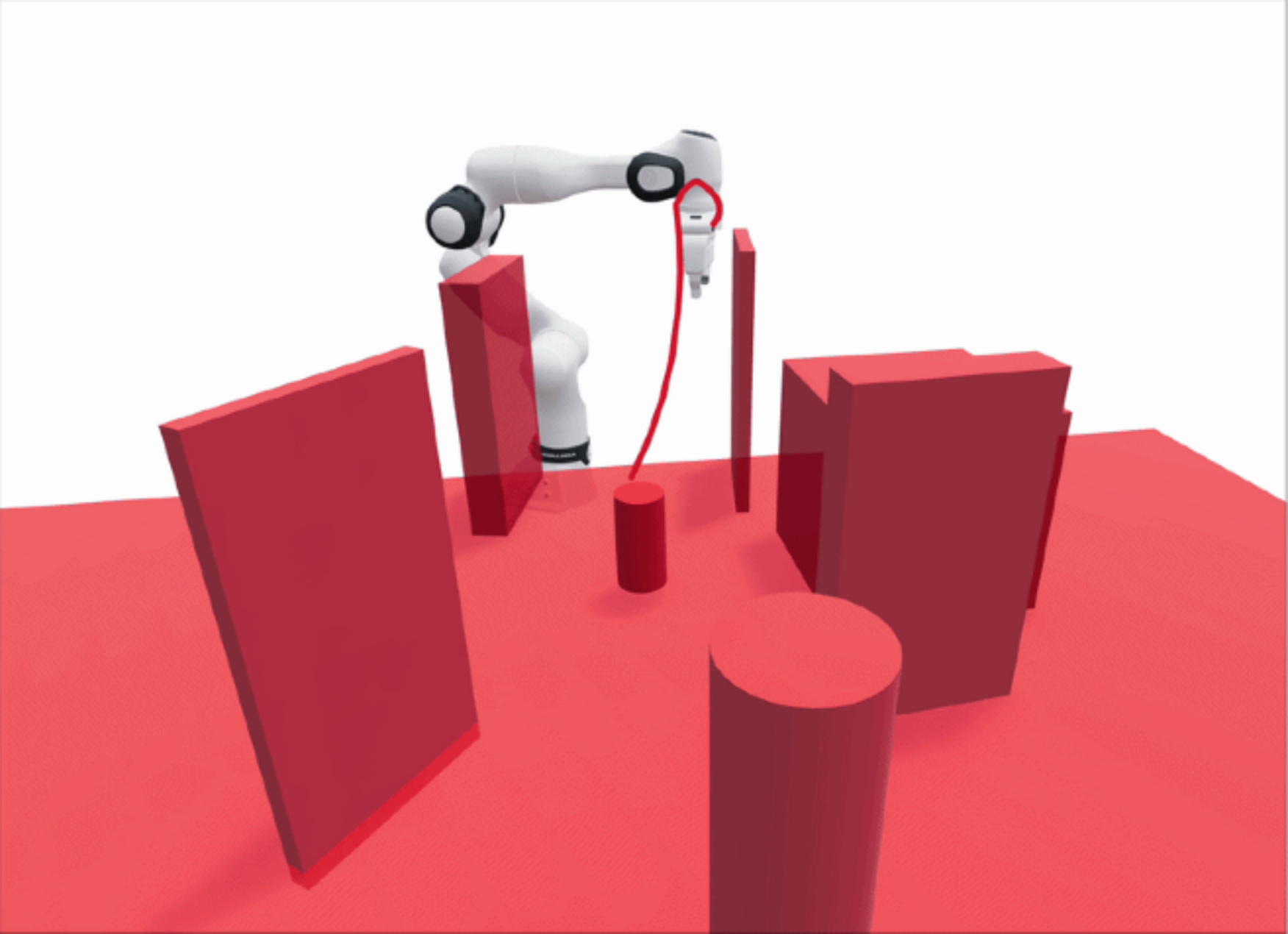
Scaling-up: the **kinetic-energy** metric produces
smoother, lower-effort motions than the Euclidean baseline

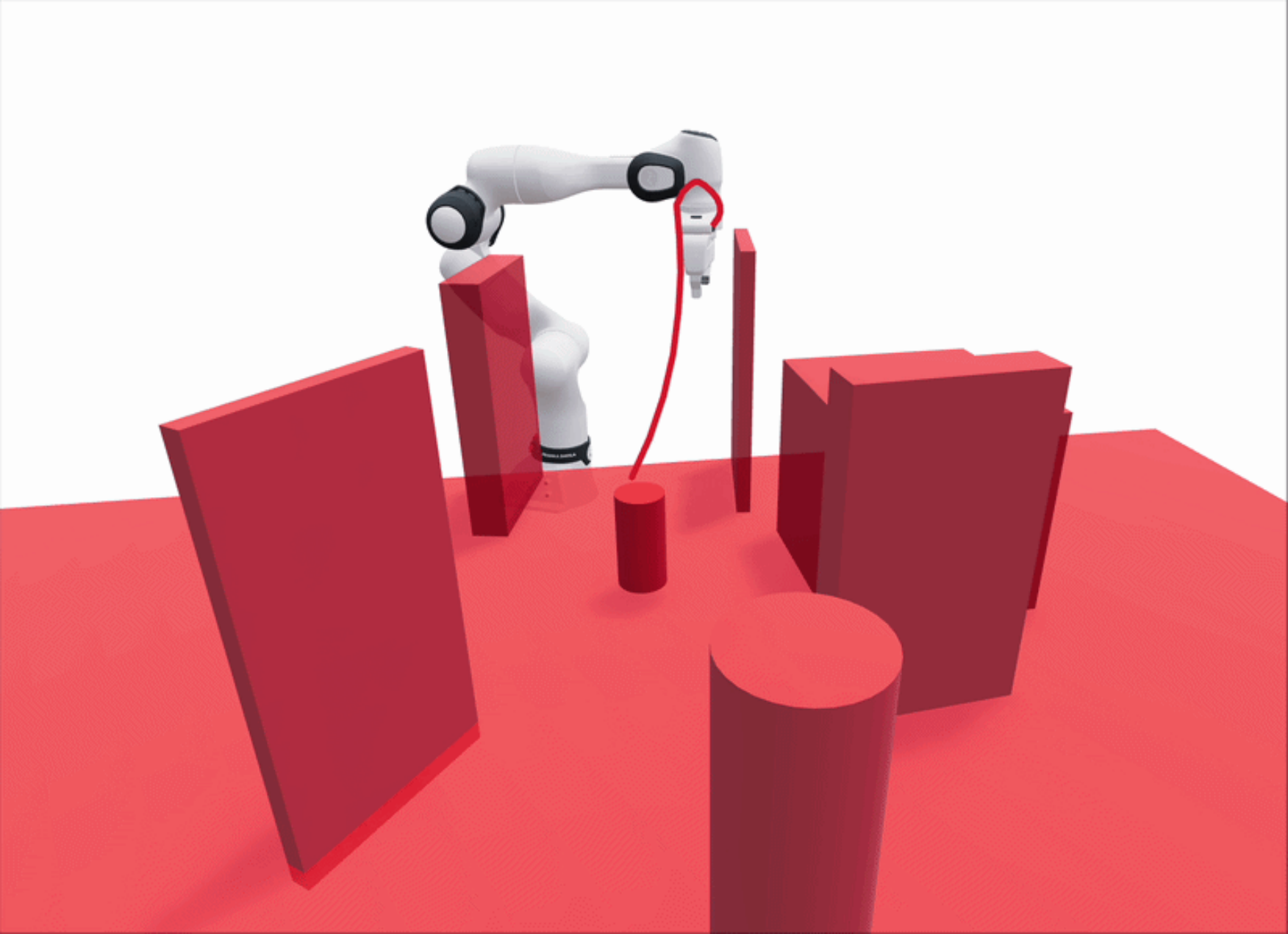
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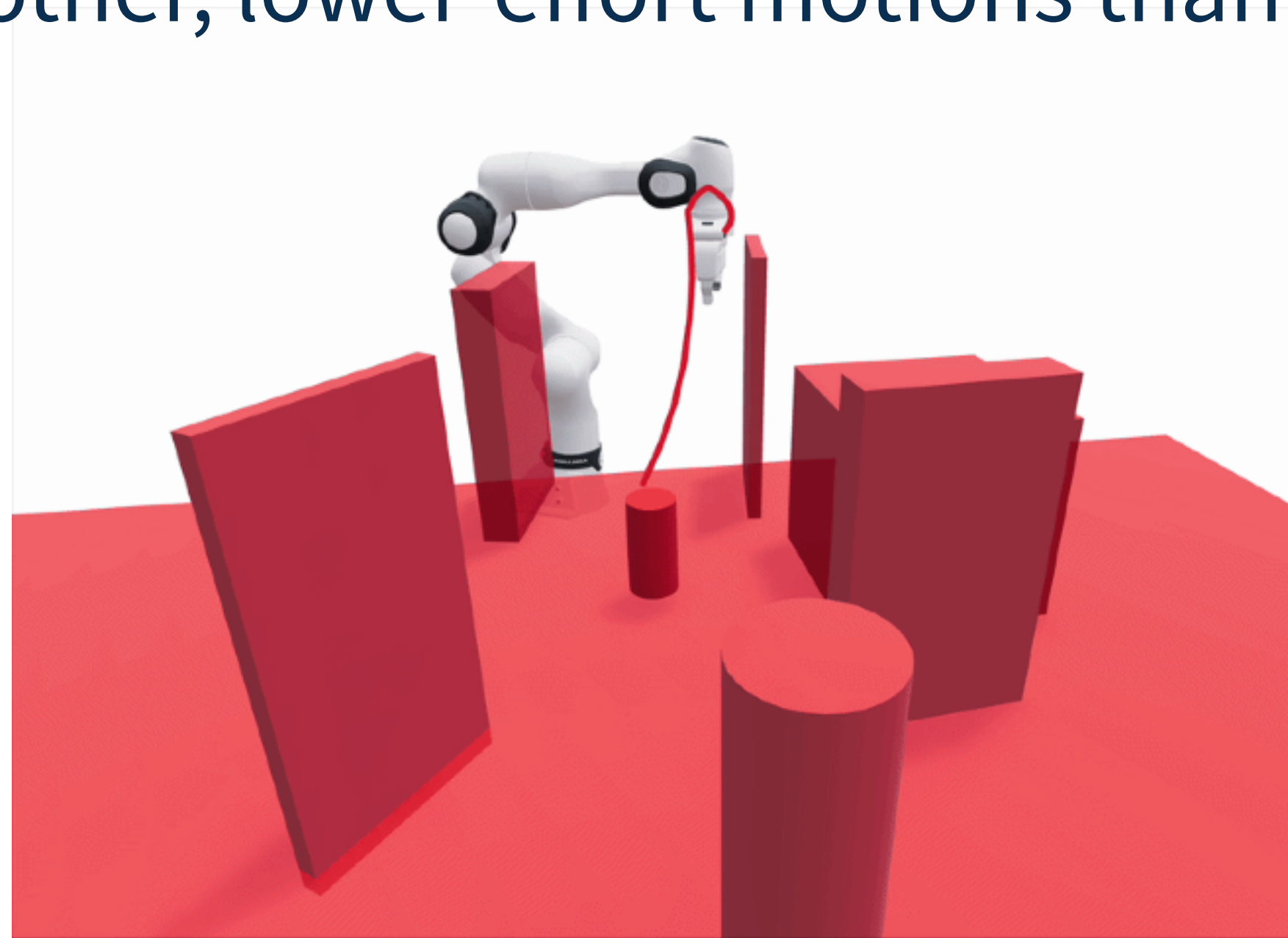






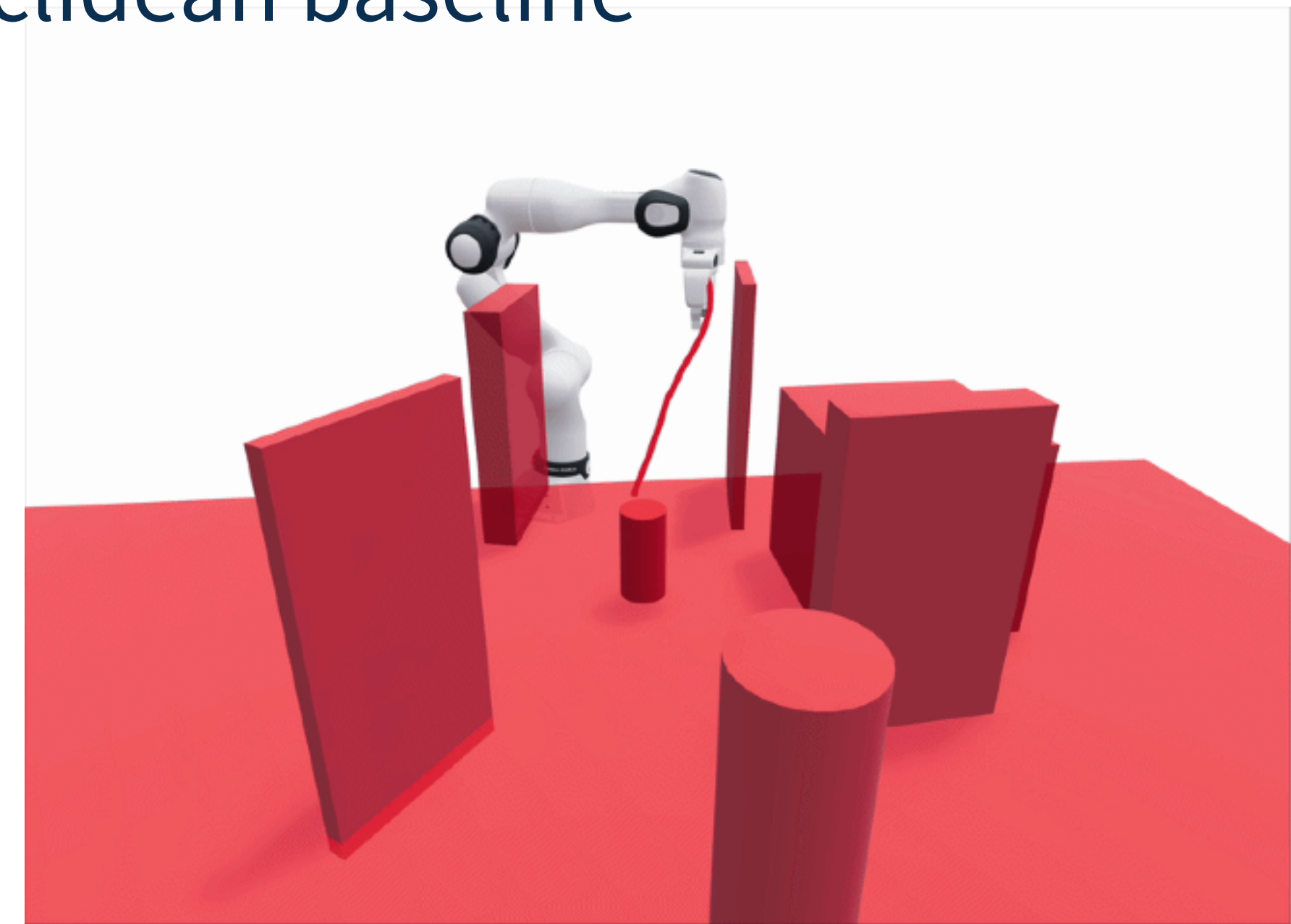


Scaling-up: the **kinetic-energy** metric produces smoother, lower-effort motions than the Euclidean baseline



Euclidean metric

joint-space straight-line motions



Kinetic-energy metric

joint-space motions that respect inertia

Optimization baselines often need hand-tuned barriers to avoid obstacles
— and can still get **stuck in local minima**.

A **left-invariant** SE(2) metric encodes (soft) non-holonomic constraints directly into distance

— we plan in *continuous* SE(2): no lattices or motion primitives are needed

$$\xi = (v_x, v_y, \omega) \in \mathbb{R}^3 \cong \mathfrak{se}(2)$$

$$\langle \xi, \xi \rangle = \xi^\top G \xi, \quad G = \text{diag}(w_x, w_y, w_\theta)$$

penalizes body-lateral motion

$$w_y \gg w_x$$

